DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs to improve slot-filling and tool call rejection without human labeling.

ML Researchers & NLP Engineers | RLHF & Tool-Augmented LLMs

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Failure Modes of SFT-Only Training

Missing Follow-ups
Fails to ask clarifying questions when user queries lack required arguments.

Hallucinated Tool Calls

Invokes tools with incomplete or fabricated slot values.

No Rejection

Cannot refuse out-of-scope requests, leading to incorrect tool invocations.

Why DPO?

SFT learns from expert trajectories but cannot distinguish good vs. bad dialogue flows.

DPO directly optimizes from paired preferences — chosen vs. rejected trajectories — without a separate reward model.

Result: better slot-filling, fewer hallucinated calls, and robust tool rejection.

Near GPT-4o performance on smaller open-source models

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

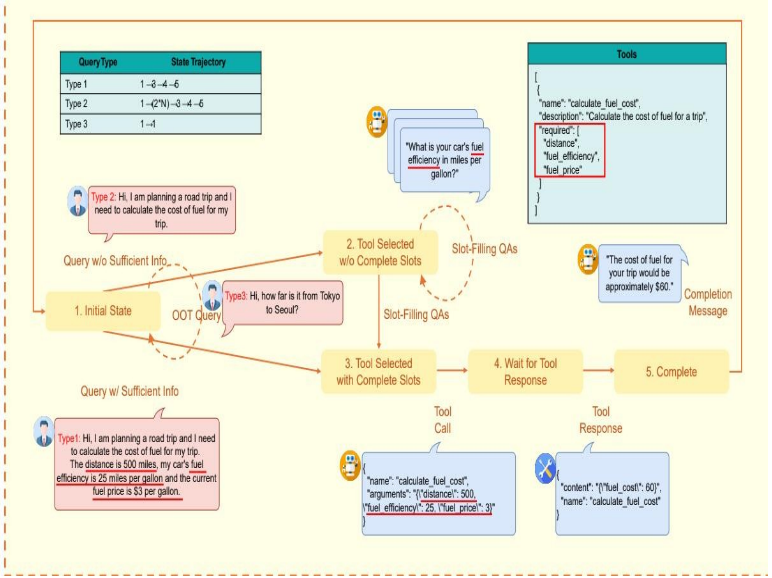


Figure 1: Five internal states and state trajectories for three query types

1 **Initial State**

No history. Out-of-scope requests return a rejection message and remain here.

2 **Tool Selected w/o Complete Slots**

Tool identified but required parameters missing. Slot-filling QA occurs here.

3 **Tool Selected w/ Complete Slots**

Both tool selection and all argument values are determined.

4 **Wait for Tool Response**

Tool call executed, awaiting results.

5 **Complete**

Tool results received; completion message generated for the user.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed.	"Check weather in Seoul today"
Type 2	1 → (2×N) → 3 → 4 → 5	Query lacks some arguments. Slot-filling QA repeats N times until all required fields are gathered.	"Book a flight" → "Where to?" → "Tokyo" → ...
Type 3	1 → 1	Requested functionality is unavailable. Tool call must be rejected.	"Translate this to Klingon" (no translation tool)

These trajectories form the basis for automatic preference pair construction — no human labeling required.

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Diff.
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089 / 562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

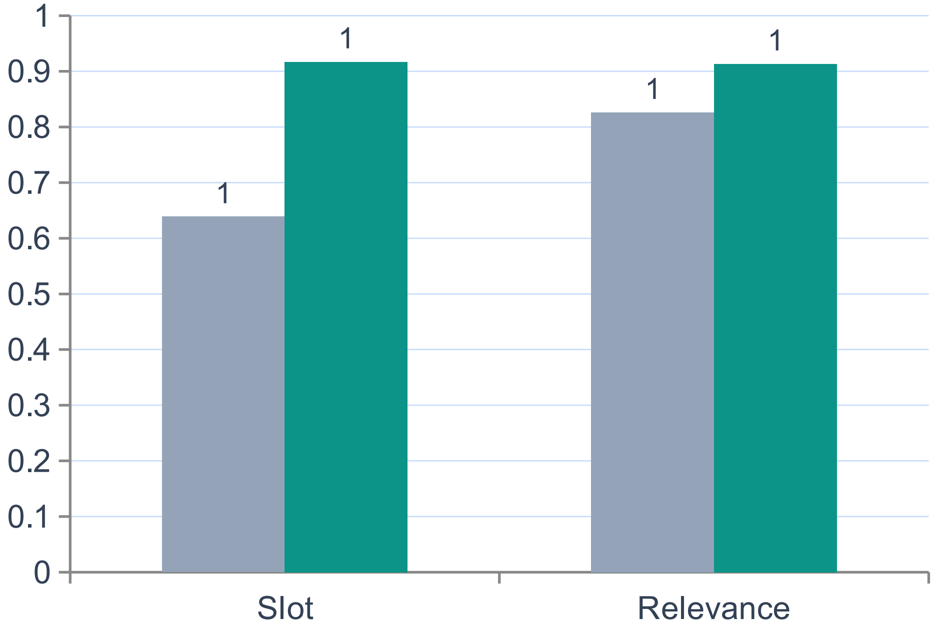
Total pairs teach specific dialogue-control lessons automatically, eliminating the need for human preference labeling.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg.	Macro Avg.
Prop.-8B	DiaTool-DPO-only	0.314	0.700	**0.833**	0.609	0.575	0.614
Prop.-8B	SFT-only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	**0.957**	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

Bold + gold = best value per column. SFT + DiaTool-DPO consistently achieves the highest Macro Average across all models.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Slot Accuracy

0.639 → 0.917

+43.5% relative improvement

Relevance

0.826 → 0.913

+10.5% relative improvement

Approaches GPT-4o: 94.8% info gathering, 91% tool rejection

Key Takeaways: RL-Based Dialogue Control for ~~Production~~ TA-LLMs

01 **MDP formulation enables systematic preference pair construction**

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

02 **DiaTool-DPO consistently improves Macro Average**

Across all three model sizes (3.1B, 8B), SFT + DiaTool-DPO achieves the best balanced performance.

03 **Slot-filling and tool rejection see the largest gains**

LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5% over SFT-only.

04 **DPO alone is insufficient**

DiaTool-DPO-only (without SFT) performs poorly, confirming that DPO complements rather than replaces supervised training.

05 **The approach is language-agnostic**

While primary experiments were in Korean, the method generalizes to English, opening broad applicability for production TA-LLMs.